

ByteTrack + LTC Motion Tracker

A research-oriented extension of ByteTrack that strengthens motion modeling with LTC/CfC predictors and adaptively balances motion and appearance cues through DMA fusion.



Robust tracking requires more than strong detection

ByteTrack provides an efficient and reliable baseline; however, its linear motion assumption becomes vulnerable under abrupt direction changes, occlusion, and dense interactions.

Established baseline

YOLOX performs frame-level detection, Kalman estimation forecasts track positions, and BYTE association links detections into consistent trajectories.

Key limitation

A one-step linear predictor cannot adequately model complex motion, while fixed motion-appearance weighting limits adaptability across scenes.

Design objective

Preserve ByteTrack's speed and simplicity while selectively improving the two highest-impact components: motion prediction and association cost fusion.

Two targeted contributions

The proposed fork introduces complementary upgrades: LTC improves non-linear motion forecasting, and DMA learns when association should rely more on motion or appearance.

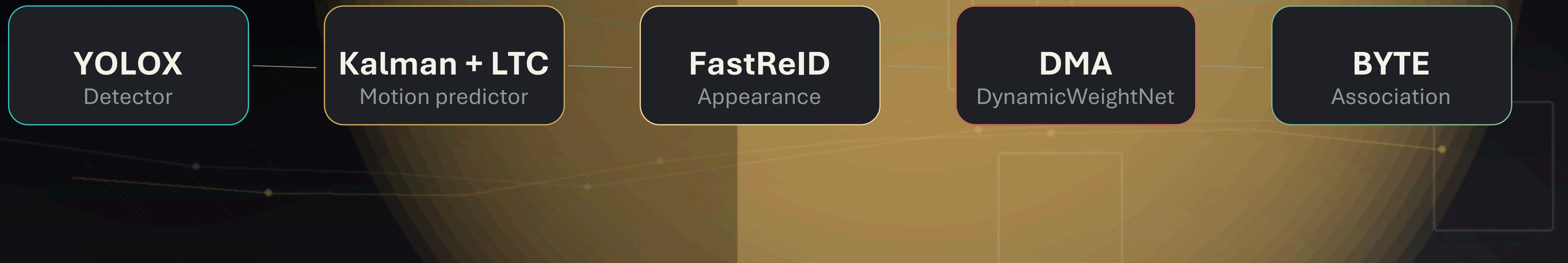
01 LTC Motion

Liquid Time-Constant / CfC models learn residual corrections over Kalman predictions, improving trajectory forecasting without destabilizing the tracker.

02 DMA Fusion

DynamicWeightNet estimates motion and ReID weights for each track-detection pair, enabling context-aware association at every frame.

Compact architecture, adaptive association



LTC Motion Predictor

The LTC predictor estimates residual motion beyond Kalman, preserving baseline stability while capturing trajectory patterns observed in real tracking data.

Liquid Time-Constant network: an ODE-inspired recurrent model with adaptive temporal dynamics.

Residual learning: complements Kalman prediction instead of replacing the baseline behavior.

Training uses MOT17 / DanceTrack ground-truth trajectories; inference is enabled through `--ltc_motion_ckpt`.

The mathematical formulation is documented in `bytetrack_math.pdf`.

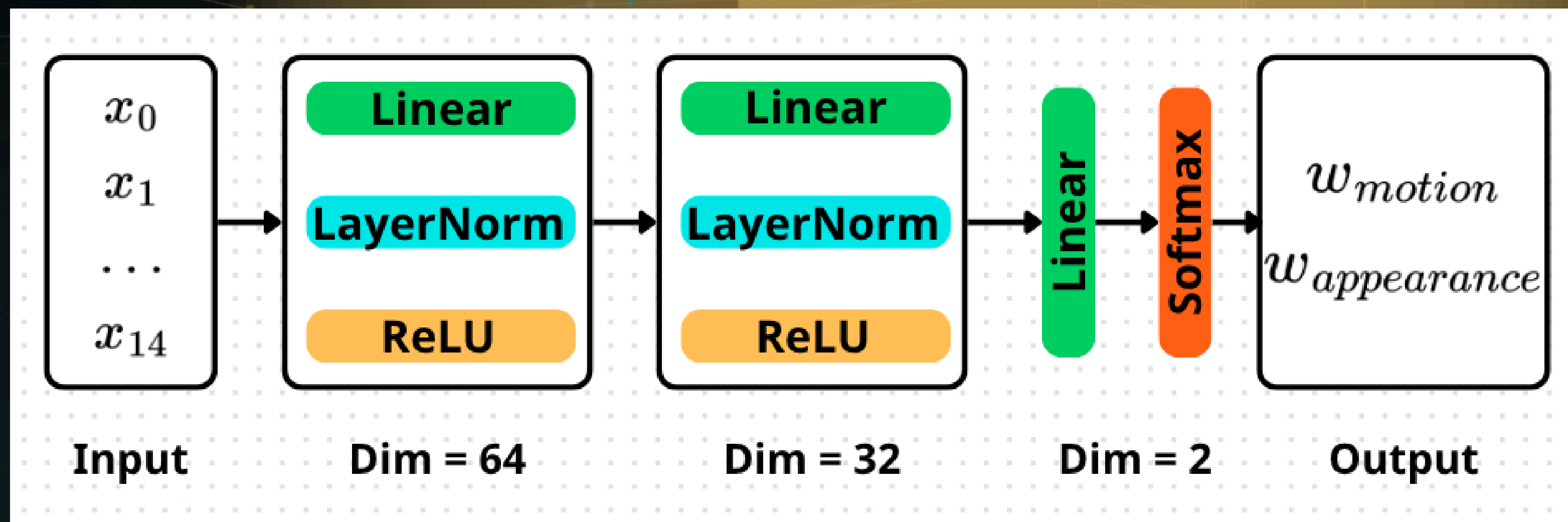
Training and inference

```
python tools/train_ltc_motion.py  
  --data_dir datasets/mot/train
```

```
python tools/track.py \  
  --ltc_motion_ckpt ltc_motion.pth
```

DMA Fusion

DMA learns to allocate trust between motion and ReID features dynamically, rather than assuming both cues are equally reliable in every scene.



DMA Fusion

Features

Motion Features

- Motion IoU
- Motion Cost
- Mahalanobis Norm
- Covariance Trace
- Covariance Mean
- Velocity Magnitude
- Time Since Update

Appearance Features

- Cosine Similarity
- Feature Variance

Detection Features

- Detection Score
- BBox Area
- BBox Aspect

Track History Features

- Track age
- Track length
- Has Appearance

DMA Fusion

Motion Features

Motion IoU: The ability to match between detection and track whether they match or not.

The larger the IoU
→ Detection and track
match each other.

$$IoU = \frac{Area_{overlap}}{Area_{union}}$$

Motion Cost: Negative correlation with IoU. The lower the cost, the greater the likelihood of a match.

$$Motion_{cost} = 1 - IoU$$

DMA Fusion

Motion Features

Mahalanobis Norm: Is the distance from a detection point to a prediction point based on covariance from the prediction.

Step 1: Transition the KF state to the measurement state.

$$\mu_z = H\mu$$

$$S = HPH^T + R$$

Step 2: Calculate the error between the detection and the KF state.

$$y = z - \mu_z$$

Step 3: Calculate Mahalanobis distance.

$$d^2 = y^T S^{-1} y$$

μ Mean state

P KF Covariance matrix

S Covariance

R KF Noise Matrix

H Observation matrix

z State detection

DMA Fusion

Motion Features

Covariance trace: The sum of the main diagonal elements of the KF covariance matrix.

Covariance mean: The average of the main diagonal elements of the KF covariance matrix.

Time since update: The number of frames matched since the last update. A lower value indicates continuous KF updates and consistently accurate IDs.

Velocity magnitude: The speed of the track's movement, indicating whether it is moving fast or slow.

$$\tanh \left(\sqrt{vx^2 + vy^2} \right)$$

DMA Fusion

Appearance Features

Cosine Similarity: The similarity between the current detection embedding and the track embedding. A higher score indicates a better match between the detection and the track.

Feature Variance: A measure of embedding stability based on the stored history of ReID embeddings. A lower score indicates higher ReID reliability.

DMA Fusion

Detection Features

Detection Score: A score derived from detection. If the score is low, both motion and appearance are considered unreliable.

BBox Area: Area of the bounding box from the detection.

BBox Aspect: Bounding box aspect ratio from detection.

DMA Fusion

Track History Features

Track Age: The total number of frames from inception to the present. A high value indicates that the track has existed for a long time and accumulated multiple matches, making it more reliable than a newly created track.

Tracklet length: The number of consecutive times the track has been successfully matched. A higher value indicates a very stable track.

Has Appearance: Has the track accumulated ReID embeddings, and do the candidate detections selected for pairing possess ReID embeddings? (No = 0, Yes = 1)

Experimental protocol

Evaluation emphasizes scenarios where linear motion models are most challenged:
non-linear trajectories, dense interactions, and visually ambiguous targets.

Datasets

MOT17 / MOT20: pedestrian tracking benchmarks
SportsMOT: non-linear motion and appearance ambiguity

Training setup

- YOLOX-X mixed detection configuration
- LTC and DMA trained on ground-truth trajectories

Evaluation criteria

TrackEval metrics: HOTA, MOTA, IDF1
Ablation compares Kalman baseline vs LTC + DMA
Research fork focused on measurable component-level gains

SportMOT benchmark

Method	HOTA	DetA	AssA	MOTA	IDF1
FairMOT	49.3%	70.2%	34.7%	86.4%	53.5%
GTR	54.5%	64.8%	45.9%	67.9%	55.8%
QDTrack	60.4%	77.5%	47.2%	90.1%	62.3%
CenterTrack	62.7%	82.1%	48.0%	90.8%	60.0%
TransTrack	68.9%	82.7%	57.5%	92.6%	71.5%
ByteTrack	62.1%	76.5%	50.5%	93.4%	69.1%
BoT-SORT	68.7%	84.4%	55.9%	94.5%	70.0%
OC-SORT	68.1%	84.8%	54.8%	93.4%	68.0%
DiffMOT	72.1%	86.0%	60.5%	94.5%	72.8%
LTC-DMA (Ours)	78.35%	88.32%	69.52%	98.12%	82.60%

78.35%

HOTA

88.32%

DetA

69.52%

AssA

82.60%

IDF1

90.32%

MOTA

MOT17 Benchmark

79.15%

DetA

77.71%

AssA

87.63%

IDF1

78.19%

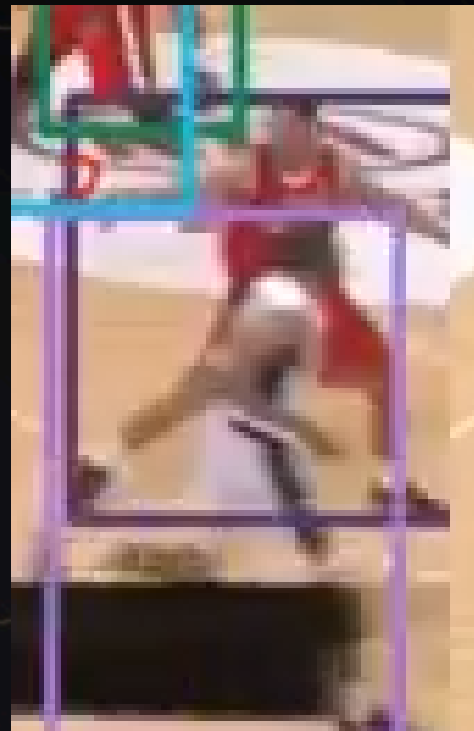
HOTA

90.32%

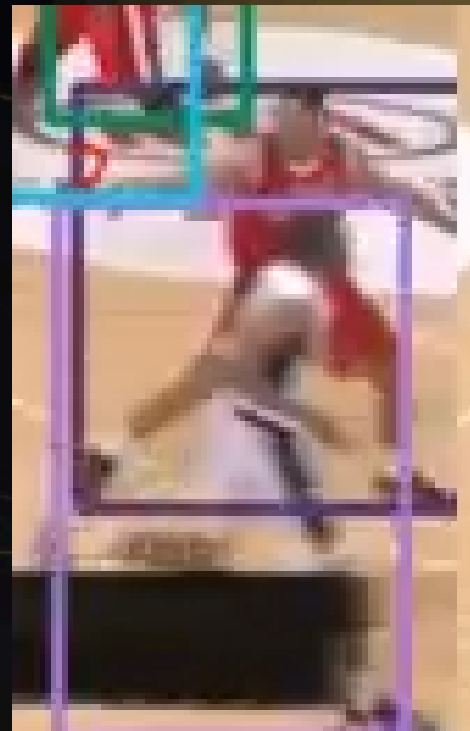
MOTA

Method	HOTA	DetA	AssA	MOTA	IDF1
DiffusionTrack	60.8%	63.2%	58.8%	77.9%	73.8%
MotionTrack	61.6%	-	60.2%	78.6%	75.1%
ByteTrack	63.1%	64.5%	62.0%	80.3%	77.3%
OC-SORT	63.2%	63.2%	63.4%	78.0%	77.5%
MixSort-OC	63.4%	63.8%	63.2%	78.9%	77.8%
MixSort-Byte	64.0%	64.1%	64.2%	79.3%	78.7%
C-BIoU	64.1%	64.8%	63.7%	79.7%	79.7%
StrongSORT	64.4%	64.6%	64.4%	79.6%	79.5%
Deep OC-SORT	64.9%	64.1%	65.9%	79.4%	80.6%
SparseTrack	65.1%	65.3%	65.1%	81.0%	80.1%
DiffMOT	64.5%	64.7%	64.6%	79.8%	79.3%
DMA (Ours)	78.19%	79.15%	77.71%	90.32%	87.63%

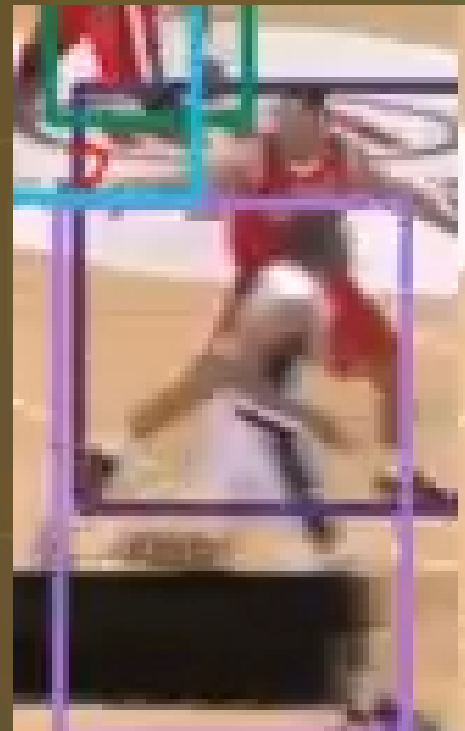
Comparasion



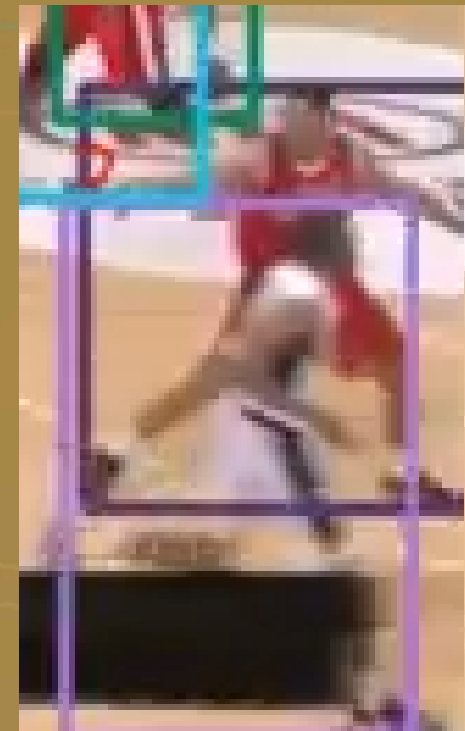
ByteTrack



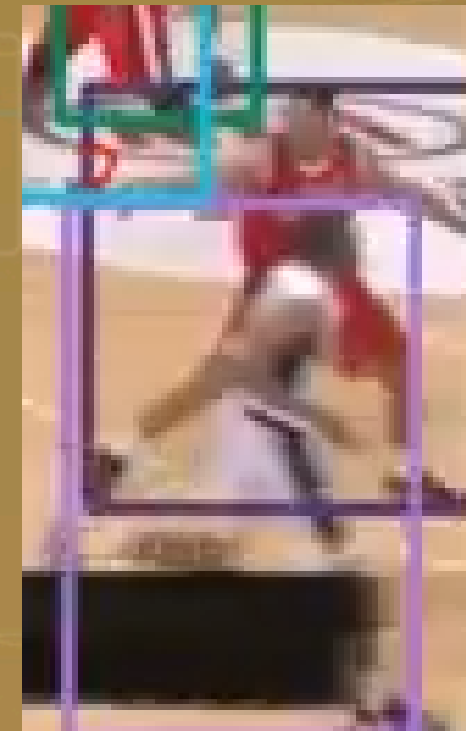
ByteTrack + ReID



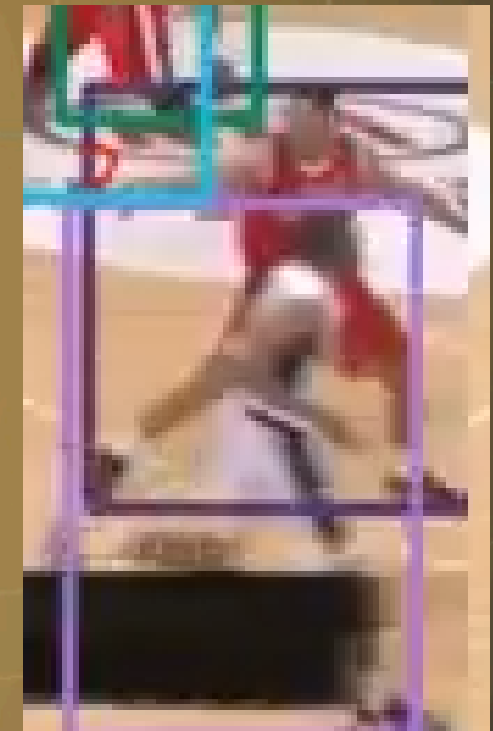
DMA



LTC



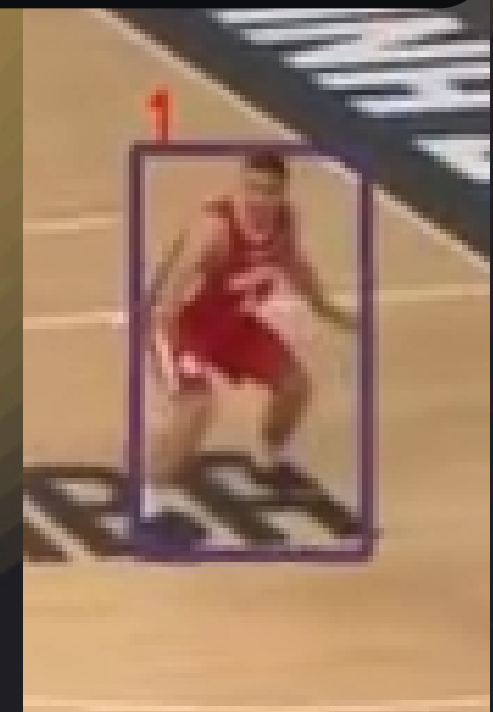
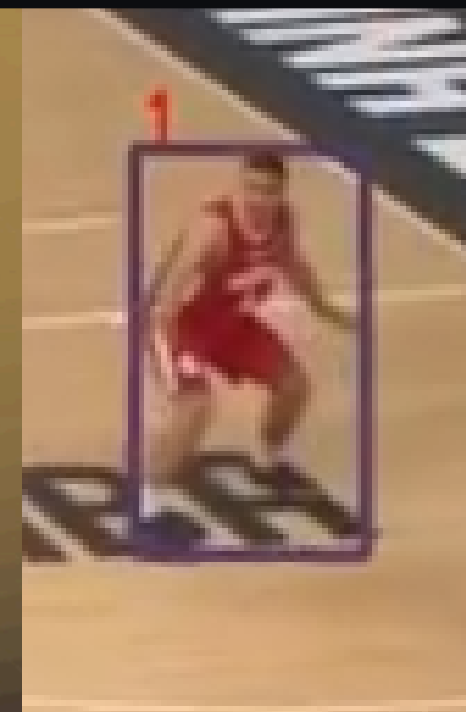
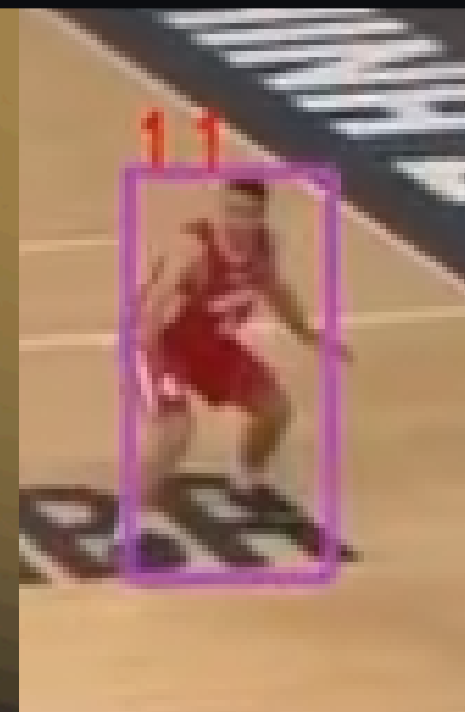
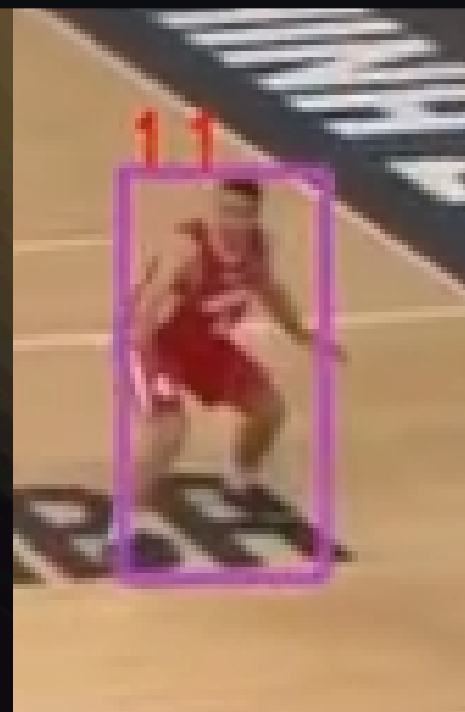
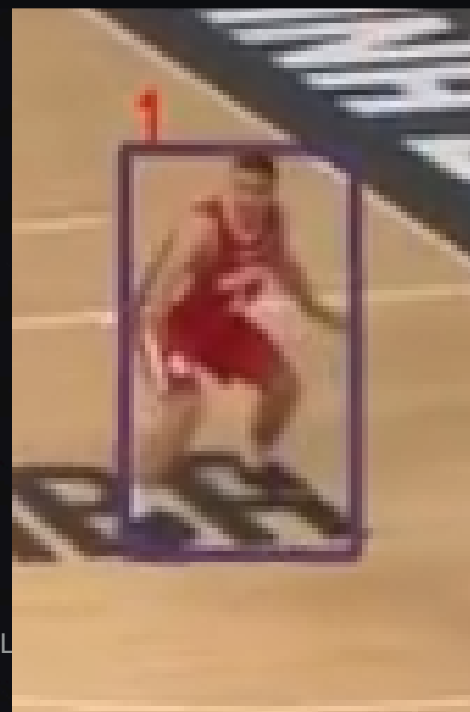
LTC + ReID



LTC + DMA



ByteTrack-DMA-L



Thank you

The proposed ByteTrack + LTC + DMA tracker improves motion robustness and adaptive association, reaching 78.35 HOTA / 98.12 MOTA on SportMOT and 78.19 HOTA / 90.32 MOTA / 87.63 IDF1 on MOT17.

[github.com/khangkaka066 · ByteTrack-DMA-LTC-Motion-Tracker](https://github.com/khangkaka066/ByteTrack-DMA-LTC-Motion-Tracker)